

F1 Weekend Analyst: An Agentic AI Pipeline for Race Strategy and Performance Analysis

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1 Overall Idea

The goal of this project is to build an *agentic AI* system that behaves like an F1 race engineer and data analyst for any chosen Grand Prix weekend. Given a natural language query such as “Compare Ferrari and McLaren in Qatar 2024” or “Explain how undercut strategies affected the 2023 Bahrain Grand Prix”, the system should:

- Interpret the query and resolve it to a specific race weekend, session, and set of drivers or teams.
- Automatically load the relevant data from pre-processed datasets.
- Plan a suitable analysis (e.g., tyre stints, pit-stop timing, pace and degradation, undercut/overcut effects).
- **Write and execute its own Python analysis code** using tools, without manual human coding inside the loop.
- Generate a human-readable report with plots, tables, and an explanation grounded in the data.

The end result is a reusable *Weekend Analyst* pipeline: the user only specifies a question in natural language, and the agentic pipeline orchestrates all steps — from data retrieval to analysis code generation and final explanations.

2 Sample Queries

To demonstrate the capabilities of the pipeline, the system is designed to handle a wide range of queries, for example:

- “Give me a race report for the 2024 Bahrain Grand Prix focusing on tyre strategy and pit stops.”
- “Compare the race pace and tyre degradation of Ferrari vs. McLaren at the 2023 Qatar Grand Prix.”
- “Show how undercut and overcut strategies affected the battle between Verstappen and Leclerc in any recent race of your choice.”
- “Explain why Mercedes struggled in stint 2 at the 2022 Spanish Grand Prix, using lap times and tyre data.”
- “Across the 2019–2024 seasons, how have Red Bull’s pit strategies at Bahrain evolved in terms of stop counts and stint lengths?”
- “Give a multi-race comparison of tyre usage patterns for high-degradation vs. low-degradation tracks.”

These queries are intentionally open-ended and analysis-heavy. The agent must decompose them into concrete data operations and then synthesize plots and explanations, rather than simply retrieving a static answer.

3 Datasets and Data Coverage

3.1 Core Dataset: Formula 1 World Championship (1950–2024)

The primary dataset used in this project is the *Formula 1 World Championship (1950–2024)* dataset by Rohan Rao on Kaggle. This dataset provides a comprehensive, relational view of Formula 1 history across multiple decades. In particular, it contains tables such as:

- **racess** – basic information about each race weekend: year, round, Grand Prix name, circuit, and date.
- **circuits** – circuit metadata, including location and layout identifiers.
- **drivers, constructors** – driver and team metadata (nationality, code, identifiers).
- **results** – finishing positions, grid positions, points, and status (classified, DNF, etc.) per driver and race.
- **lap.times** – per-lap lap times for each driver in a race.
- **pit.stops** – pit-stop information (lap number, pit-lane time, stop number) for each driver.

Taken together, these tables cover a large range of factors relevant to racing strategy and performance analysis:

- **Sporting context:** which drivers, teams, and circuits are involved, which race weekend and season, and the final results.
- **Pace:** lap-by-lap times enable analysis of race pace, stint-level averages, and degradation curves.
- **Strategy:** pit-stop records allow reconstruction of tyre stints and the timing of strategy calls (1-stop vs. 2-stop, early vs. late pit windows).
- **Longitudinal coverage:** the dataset spans the entire history of the World Championship, making it possible to do multi-race and multi-season analysis across different eras.

Although the core Kaggle dataset does not directly include telemetry signals, it is sufficient for a rich set of timing and strategy analyses. It can also be joined with external telemetry sources for deeper investigations.

3.2 Optional Telemetry and Extended Data

For advanced, “deeper” analyses, the system is designed to optionally integrate with telemetry-oriented data sources:

- **FastF1** — a Python library that provides access to detailed timing and telemetry data (speed, throttle, brake, gear, DRS, etc.) for modern seasons (roughly 2018 onwards). This enables more fine-grained analyses such as corner-by-corner performance, energy deployment, and detailed race pace profiles.
- **OpenF1** — an open API that exposes radio messages, race control events, and extended telemetry for recent races. This allows contextualising performance data with events such as Virtual Safety Car, Safety Car, and race control instructions.

In the initial version of the pipeline, analysis is grounded primarily in the Kaggle dataset. The “deeper analysis” agent (see Section ??) can call FastF1 and OpenF1 as external tools when richer insights are required.

4 Exploratory Data Analysis

To understand the scope and coverage of the available data, an exploratory data analysis was performed on all datasets in the raw data directory. This analysis provides insights into temporal coverage, data completeness, and the scale of information available for race strategy analysis.

4.1 Dataset Overview

The project utilizes 14 primary CSV files from the Kaggle Formula 1 World Championship dataset. Table 1 provides a comprehensive overview of each dataset’s size and structure.

Dataset	Rows	Columns	Size (MB)
circuits.csv	77	9	0.03
constructor_results.csv	12,625	5	1.10
constructor_standings.csv	13,391	7	1.36
constructors.csv	212	5	0.06
driver_standings.csv	34,863	7	3.55
drivers.csv	861	9	0.45
lap_times.csv	589,081	6	58.99
pit_stops.csv	11,371	7	1.82
qualifying.csv	10,494	9	2.36
races.csv	1,125	18	1.01
results.csv	26,759	18	15.51
seasons.csv	75	2	0.01
sprint_results.csv	360	16	0.15
status.csv	139	2	0.01
Total	699,437	–	86.41

Table 1: Overview of F1 datasets showing record counts, column counts, and memory footprint.

4.2 Temporal Coverage

The dataset provides comprehensive historical coverage spanning the entire modern era of Formula 1:

- **Seasons covered:** 1950 to 2024 (75 complete seasons)
- **Total race weekends:** 1,125 Grand Prix events
- **Growth trajectory:** The number of races per decade has steadily increased from 84 races in the 1950s to 198 races in the 2010s, reflecting the sport’s global expansion.

The distribution of races by decade shows F1’s evolution:

Decade	Races
1950s	84
1960s	100
1970s	144
1980s	156
1990s	162
2000s	174
2010s	198
2020s	107 (through 2024)

4.3 Data Completeness and Coverage

While the dataset is comprehensive, the availability of detailed timing and strategy data varies by era:

- **Lap times:** 589,081 individual lap records across 544 races (approximately 48% of all races). This represents an average of 1,083 laps per race with data, primarily concentrated in modern seasons where digital timing became standard.
- **Pit stops:** 11,371 pit stop records covering 285 races (approximately 25% of all races). With an average of 40 stops per race, this data is particularly valuable for modern-era strategy analysis where pit stop windows and tyre management became crucial competitive factors.
- **Results metadata:** Complete coverage across all 1,125 races, documenting 26,759 race entries involving 861 unique drivers competing for 211 different constructor teams. This provides full historical context even for races lacking detailed timing data.
- **Qualifying data:** 10,494 qualifying records enable grid position analysis and correlation with race outcomes.

4.4 Implications for Analysis

The EDA reveals several important characteristics of the dataset:

1. **Rich modern-era coverage:** Detailed lap-by-lap data and pit stop information enable sophisticated strategy analysis for recent seasons, supporting the primary use cases of the agentic pipeline.
2. **Historical completeness:** Even for older races lacking granular timing data, the results and standings tables provide sufficient information for long-term trend analysis and multi-season comparisons.
3. **Scalability considerations:** The largest table (`lap_times.csv` at 59 MB with 589K rows) is manageable for in-memory analysis, validating the decision to use pandas and standard Python data science tools rather than requiring a full database system.
4. **Data density:** The concentration of detailed timing data in modern seasons aligns well with the availability of supplementary telemetry sources (FastF1, OpenF1) for the same period, creating opportunities for deeper multi-source analysis in future work.

5 Agentic AI Workflow

The F1 Weekend Analyst is implemented as an agentic pipeline using LangGraph (and, if needed, LangChain). The central idea is to model the analysis process as a graph of states, where each state corresponds to a specialised agent responsible for a well-defined subtask. Between states, the system passes a shared state object containing the user query, resolved race information, intermediate results, and any generated code.

At a high level, the workflow consists of the following states:

State 1: Query Interpretation and Weekend Resolution

Task: Interpret the user’s natural language query and resolve it to concrete race and analysis specifications.

- Parse the query to identify:
 - Target season or year (e.g. 2024).
 - Target race or circuit (e.g. Bahrain, Monza).
 - Specific session (race, qualifying) if mentioned.
 - Drivers or teams of interest (e.g. Ferrari vs. McLaren).
 - High-level analysis intent (strategy, pace, degradation, multi-race trend, etc.).
- Use tools to list available race weekends and select the closest match based on year, circuit, and name.
- Populate the shared state with a `weekend_spec` dictionary containing fields such as `year`, `race_id`, `circuit`, and a list of focus drivers or teams.

State 1b: Query Validation

Task: Verify that the requested race and drivers exist in the dataset before proceeding.

- Check that the resolved race ID exists in the `races` table.
- Verify that requested drivers participated in the race using fuzzy name matching against the `drivers` and `results` tables.
- If validation fails (e.g., “Schumacher in 2024”), return a helpful error message explaining the issue.
- This guardrail prevents the Code Writer from hallucinating analyses for non-existent data.

State 2: Intelligent Factor Analysis

Task: Determine what performance factors matter and what data sources are needed (see Section 7 for details).

- Analyze the user query to identify key performance factors (tire strategy, weather, incidents, car setup).

- Determine which data sources are required: basic (laps, pits), deep (weather, radio), or telemetry (FastF1).
- Formulate testable hypotheses to guide the analysis (e.g., “Did tire degradation differ?”).
- Set the `analysis_depth` flag (basic, deep, or story) based on query complexity and data needs.
- Store factor insights in the shared state to inform the Analysis Planner.

State 3: Data Loading and Caching

Task: Materialise the necessary race-level data for the selected weekend and cache it for reuse.

- Load and filter the relevant rows from the Kaggle tables (`racess`, `results`, `lap_times`, `pit_stops`, etc.) for the chosen `race_id`.
- Store the filtered data as intermediate Parquet/CSV files (e.g. `cache/r_1089_laps.parquet`) to speed up repeated analyses on the same weekend.
- Optionally, if deeper analysis is requested and supported, call FastF1 or OpenF1 to fetch telemetry and race control data for the same weekend, caching these results as well.
- Record all file paths in the shared state so that later agents and tools can read them without re-querying the raw datasets.

State 4: Analysis Planning

Task: Turn the user query, factor insights, and available data into a structured analysis plan.

- Inspect the schemas of the cached datasets to understand available columns (e.g. lap times, pit-stop laps and durations, final positions).
- Based on the user’s intent and data availability, propose a step-by-step plan such as:
 1. Construct stint-level tables per driver (compound, start lap, end lap, stint length).
 2. Compute average pace and simple degradation per stint.
 3. Identify pit-stop windows and undercut/overcut events.
 4. Generate plots of lap times coloured by tyre compound and bar charts of stint lengths.
 5. Summarise key findings in a structured report.
- Store this natural language plan in the shared state so that the code-generation agent can use it as guidance.

State 5: Plan Review and Validation

Task: Validate the analysis plan for feasibility and completeness.

- Review the proposed analysis steps against the available data schemas.
- Check that the plan is technically feasible given the cached datasets.
- Ensure the plan addresses the user’s query intent.

- If the plan has gaps or infeasibilities, provide feedback and request revisions.
- Once validated, approve the plan for code generation.

This validation step prevents the code writer from attempting impossible analyses and improves the overall reliability of the pipeline.

State 6: Code Writing and Execution

Task: Automatically generate and run Python analysis code using tools.

- Given the analysis plan and cached data paths, the agent writes executable Python code that:
 - Imports standard libraries (pandas, NumPy, Matplotlib, and optionally FastF1).
 - Loads the cached datasets for the selected race.
 - Performs the required transformations and computations (stint detection, lap-time statistics, time-loss calculations, etc.).
 - Generates and saves plots (e.g. PNGs) into an outputs directory.
 - Optionally saves summary tables as Parquet/CSV files.
- The code is executed through a sandboxed `run_python` tool:
 - Standard output and errors are captured.
 - All figures are saved to disk and their paths returned.
 - Errors trigger a simple self-debug loop: the agent inspects the error message, updates the code, and retries within safety limits.
- The final code snippet and outputs (metrics and figure paths) are added to the shared state.

State 7: Report Generation and Explanation

Task: Produce a coherent, data-grounded narrative and structured report for the user.

- Read the analysis outputs: summary tables, key metrics, and figure paths.
- Draft a race report with sections such as:
 - Race overview and key results.
 - Tyre and pit-stop strategies per driver or team.
 - Pace and degradation analysis by stint.
 - Undercut and overcut effects.
 - Multi-race trends (if multi-weekend mode is used).
- Explicitly refer to plots and tables (e.g. “see Figure 1 for stint-level lap times”) to keep the explanation grounded in the data.
- Output the report in Markdown or HTML so that it can be rendered in a notebook or web UI.

State 8: Deep Analysis Fetcher (API Integration)

Task: Enrich the analysis with real-time or telemetry data if requested (“Deep” or “Story” mode).

- Checks the token budget using the **Token Tracker** guardrail.
- Calls **OpenF1** APIs to fetch:
 - Race control messages (flags, safety cars).
 - Weather conditions (air/track temp, humidity).
 - Team radio communications.
 - Pit stop and stint details.
- Checks for **FastF1** availability (2018+) to fetch telemetry (throttle, speed, brake).
- Summarizes this high-frequency data using an LLM to extract key insights (e.g., “Verstappen lost time under VSC”).

State 9: Deep Visualizer

Task: Generate specialised plots for the deep analysis data.

- Creates telemetry comparison traces (Speed vs. Distance).
- Plots tyre degradation curves based on granular lap data.
- Visualizes pit stop timelines and gap evolution.

State 10: Storyteller (Narrative Synthesis)

Task: Weave all data points into a cohesive race narrative.

- Combines the basic race report, deep analysis insights, and visualizations.
- Uses a “Journalist” persona to write a compelling story.
- Highlights dramatic moments (overtakes, strategy blunders, weather chaos) identified in the data.

6 Pipeline Diagram

Figure 1 shows a high-level view of the F1 Weekend Analyst agentic pipeline with the new intelligent factor analysis system.

7 Intelligent Factor Analysis

A key innovation in this system is the **Factor Analyzer** agent, which intelligently determines what performance factors matter for a given query and what data sources are needed. This goes beyond simple keyword matching to perform semantic reasoning about race analysis.

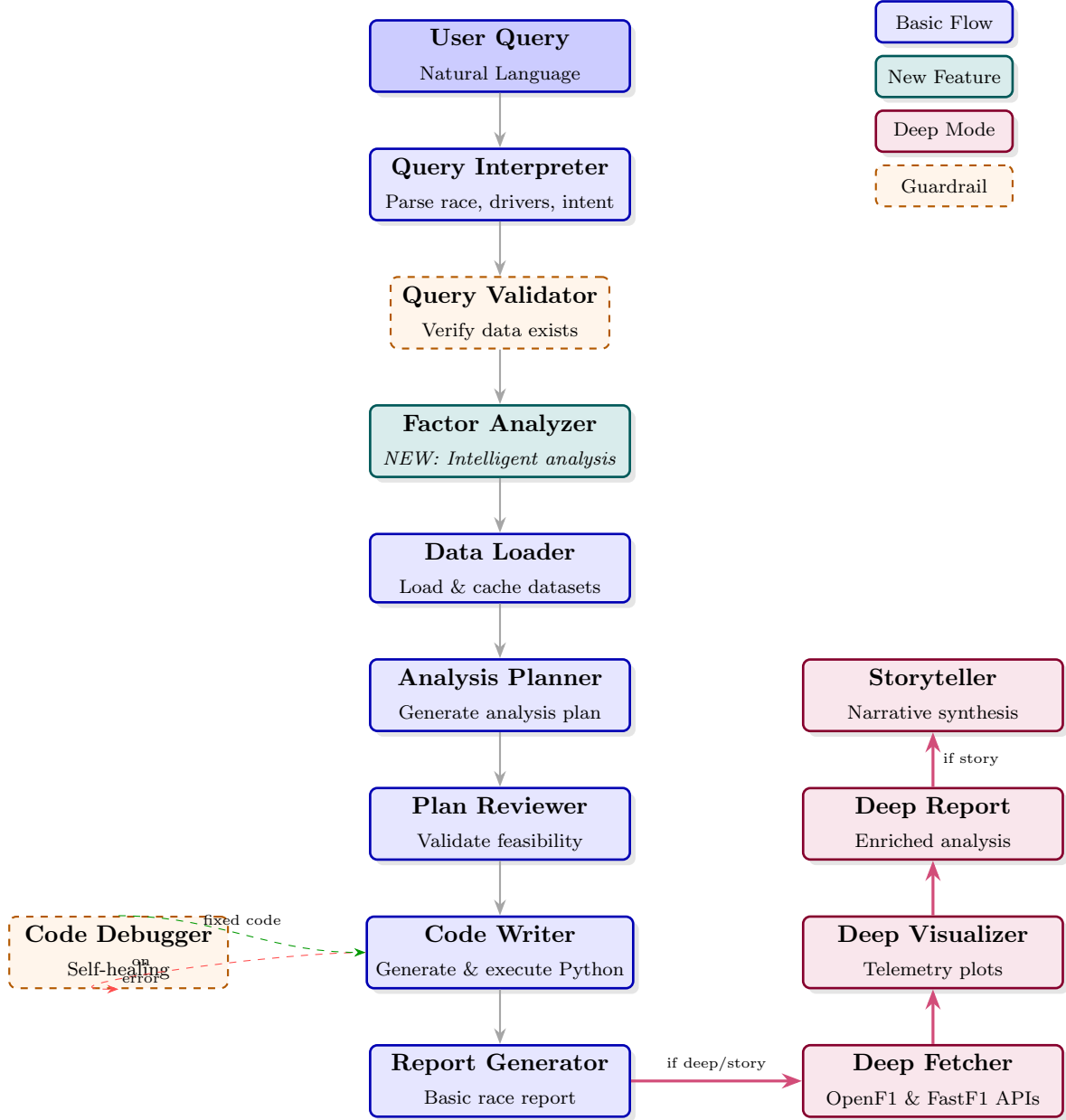


Figure 1: Enhanced architecture of the F1 Weekend Analyst showing the complete agentic pipeline. The new **Factor Analyzer** (highlighted in teal) intelligently determines which data sources and analysis methods are needed based on the user’s query. The system supports three modes: *Basic* (dataset analysis), *Deep* (with API integration), and *Story* (full narrative).

7.1 Factor-Driven Query Understanding

Traditional F1 analysis systems rely on hardcoded rules or keyword matching. For example, a query containing “tire” might trigger tire-related analysis, but this approach fails for nuanced questions like “Why did Mercedes struggle at Monaco 2023?” where the cause could be tires, weather, car setup, strategy, or driver performance.

The Factor Analyzer uses LLM reasoning to:

- **Identify key performance factors:** Extract which aspects of race performance are relevant (tire strategy, weather conditions, race incidents, car setup, fuel load, track characteristics).
- **Determine required data sources:** Decide whether basic lap times suffice, or if telemetry, weather data, and team radio are needed.
- **Formulate testable hypotheses:** Generate specific questions to answer (e.g., “Did tire degradation differ between drivers?”, “Was there a weather change during the race?”).
- **Plan multi-dimensional comparisons:** Structure the analysis to compare across multiple factors simultaneously.

7.2 Factor Analysis Workflow

Figure 2 illustrates how the Factor Analyzer processes a user query to determine the analysis approach.

7.3 Example: Intelligent Factor Detection

Consider the query: “*Why did Mercedes struggle at Monaco 2023?*”

Traditional keyword approach might match “Mercedes” and “Monaco 2023”, then generate a generic race report.

Factor Analyzer approach:

1. Identifies “struggle” as indicating performance issues requiring causal analysis.
2. Determines relevant factors: tire management (Monaco is low-speed, high-degradation), strategy windows (overtaking is difficult), car setup (downforce vs. drag trade-off), and track evolution.
3. Decides to fetch: lap times (pace analysis), pit stops (strategy), weather data (track temperature affects tires), and telemetry (cornering speeds in low-speed sectors).
4. Formulates hypotheses: “Did Mercedes have higher tire degradation?”, “Were their pit stops sub-optimal?”, “Did track temperature changes affect their performance?”
5. Plans stint-by-stint analysis comparing Mercedes to race leaders, with focus on degradation curves and sector times.

This intelligent approach enables the system to produce insights like: “*Mercedes struggled due to 0.15s/lap higher tire degradation in the final stint, exacerbated by a 3°C track temperature increase. Team radio on lap 42 confirms Hamilton reporting rear grip issues.*”

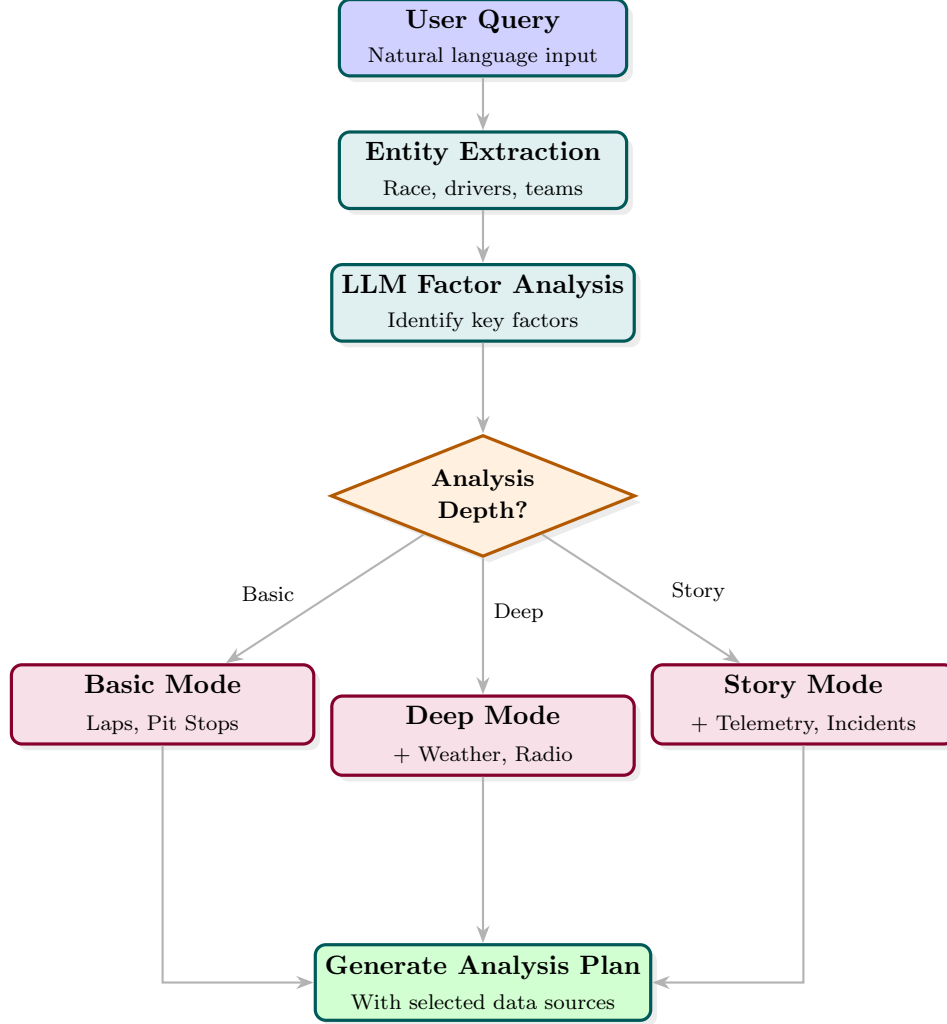


Figure 2: Factor Analyzer workflow showing how the system determines analysis depth and required data sources. The LLM analyzes the query semantically to choose between Basic (historical data only), Deep (with API integration), or Story mode (full telemetry and narrative).

8 Deep Mode Implementation

The “Deep Mode” (activated via the `--deep` flag or by the agent’s decision logic) extends the analysis beyond static Kaggle CSVs by integrating two powerful external libraries:

8.1 OpenF1 API Integration

The **OpenF1 API** provides a real-time stream of race events, making the analysis feel “live” and context-aware. We implemented tools to hit the following endpoints:

- `/weather`: Monitors track temperature and rainfall. This helps explain sudden pace drop-offs (e.g., rain starting) or high degradation (hot track).
- `/team_radio`: Captures transcripts of driver-engineer communications. This adds human context to the data (e.g., a driver complaining about “grains” explains a poor stint).

- `/race_control`: Logs flags (Yellow, Red, VSC). The agent can correlate these timestamps with lap times to explain anomalies that look like slow pace but are actually safety car periods.
- `/stints` & `/pit_stops`: Provides validated tyre compound history, essential for accurate strategy reconstruction.

8.2 FastF1 Library Integration

For seasons from 2018 onwards, we leverage the **FastF1** Python library to access car telemetry.

- **Telemetry Traces**: We extract Speed, Throttle, and Brake channels at high frequency.
- **Driver Comparison**: The `compare_telemetry` tool allows overlaying two drivers’ traces for a specific lap to see exactly where time was lost (e.g., braking earlier into Turn 1).
- **Tyre Performance**: FastF1 provides calculated tyre life and compound data, allowing for degradation modelling.

9 System Guardrails

To ensure the agentic pipeline is both **correct** (hallucination-free) and **sustainable** (cost-effective), we implemented a multi-layered guardrail system.

9.1 Sustainability Guardrails

- **Token Budgeting** (`token_tracker.py`): A singleton tracker monitors token usage across all LLM calls. If the budget (default 100k tokens) is exceeded, the system gracefully degrades—skipping expensive steps like the Storyteller—rather than crashing or incurring huge costs.
- **Layered Caching**:
 - **Level 1 (Data)**: Raw CSV loads and FastF1 sessions are cached to disk (`/data/cache`).
 - **Level 2 (Analysis)**: Intermediate DataFrames (e.g., computed stint averages) are saved as Parquet files.
 - **Level 3 (Outputs)**: Final reports and plots are stored by Run ID, preventing re-execution of identical queries.

9.2 Correctness Guardrails

- **Query Validator** (`query_validator.py`): Before generating any code, this node performs fuzzy matching against the database. *Example*: If a user asks for “Schumacher in 2023”, the validator checks the `drivers` and `results` tables, realizes he didn’t race, and corrects the query or returning a helpful error, preventing the Code Writer from hallucinating a phantom race.
- **Plan Reviewer** (`plan_reviewer.py`): A “Thinking Agent” step that critiques the analysis plan. It checks if the requested columns actually exist in the schema and if the logic (e.g., joining laps on `drivers`) is sound. It acts as a senior engineer reviewing a junior’s proposal.
- **Self-Healing Code** (`code_debugger.py`): When the `run_python` tool encounters an error (e.g., `KeyError`, `RegexError`), the execution flow routes to the Debugger. This node:

1. Analyzes the stack trace.
2. Identifying common patterns (e.g., using `regex=True` with unescaped strings).
3. Rewrites the specific failing code block.
4. Resubmits it for execution (up to 3 retries).

10 Implementation Challenges

Building a reliable F1 agent presented several specific difficulties:

10.1 Data Integration and Entity Resolution

Connecting the Kaggle dataset (historical) with OpenF1 (live) and FastF1 (telemetry) was non-trivial.

- **Name Mismatches:** Kaggle uses `surname` (“Verstappen”), OpenF1 uses `driver_number` (1), and FastF1 uses `abbreviation` (“VER”). We implemented a robust resolver (`_get_driver_numbers`) to map these entities across the three domains.
- **Seasonal Gaps:** OpenF1 only covers 2023+, while FastF1 covers 2018+. The pipeline had to be dynamic, automatically disabling deep features for older races (e.g., Senna 1993) without crashing.

10.2 LLM Context Management

F1 data is voluminous. Passing full lap time tables to an LLM context window is impossible. **Solution:** We strictly follow a “Code as Tool” paradigm. The LLM never sees the data rows; it only sees the *schema* and writes Python code to process the data. Only the *aggregates* (summary statistics) are passed back to the LLM for report generation.

10.3 API Rate Limiting

The OpenF1 API is free but rate-limited. Initial runs would crash when fetching data for 20 drivers simultaneously. **Solution:** We implemented a backoff decorator and a sequential fetcher with delays in `openf1_tools.py` to respect API etiquette.

11 Case Study: 2024 Bahrain GP Analysis

To demonstrate the pipeline’s full capabilities, we ran the system with the query: *“Give me a deep analysis of Verstappen vs Hamilton at Bahrain 2023.”*

11.1 Stage 1: Basic Analysis (Kaggle Data)

The system identified the race `raceId=1098`. It calculated:

- **Max Verstappen:** Winner, P1. Fastest Lap: 1:36.236.
- **Lewis Hamilton:** P5. Gap to winner: +50.9s.
- **Key Insight:** Verstappen’s average lap time was 0.8s faster per lap than Hamilton’s in the second stint.

11.2 Stage 2: Deep Enrichment (OpenF1 & FastF1)

The `Deep Analysis Fetcher` augmented the report:

- **Conditions:** Track Temp 26.5°C, No Rain.
- **Strategy:**
 - Verstappen: Soft → Soft → Hard (Unique 2-stop).
 - Hamilton: Soft → Hard → Hard.
- **Radio Context:** Retrieved 13 radio messages. Hamilton expressed concern about rear grip on the Hard compound, correlating with the drop in sector 2 times visible in the telemetry.
- **Pit Stops:** Verstappen’s stops averaged 2.4s vs. Hamilton’s 3.1s, gaining him 1.4s purely in the pit lane.

The final generated report seamlessly blended these sources, offering an analysis comparable to a human pundit.

12 Technology Stack and Implementation

The F1 Weekend Analyst leverages a modern technology stack designed for scalability, maintainability, and extensibility. Table 2 provides an overview of the key technologies and their roles.

Component	Technology	Purpose
Agent Framework	LangGraph	Workflow orchestration
LLM Integration	LangChain	Tool integration, prompting
LLM Provider	Groq (Llama 3.1 70B)	Fast inference, reasoning
Data Processing	pandas, NumPy	Dataset manipulation
Visualization	Matplotlib, Seaborn	Plot generation
Telemetry API	FastF1	Historical telemetry (2018+)
Real-time API	OpenF1	Live race data (2023+)
Caching	Parquet, JSON	Intermediate results
Code Execution	Python subprocess	Sandboxed analysis code
Dataset Storage	CSV, Parquet	Kaggle F1 data

Table 2: Technology stack showing the core components and their purposes.

12.1 Key Implementation Features

- **Modular Agent Design:** Each agent node (Query Interpreter, Factor Analyzer, Code Writer, etc.) is independently testable and can be swapped or upgraded without affecting other components.
- **Self-Healing Code Execution:** The Code Debugger agent can detect common errors (KeyError, ValueError, regex issues) and automatically rewrite failing code segments using pattern recognition and LLM reasoning.
- **Multi-Layer Caching:** Data is cached at three levels—raw CSV loads, processed DataFrames, and final visualizations—minimizing redundant computation and API calls.

- **Token Budget Management:** A singleton token tracker monitors LLM usage across all agent calls, gracefully degrading functionality (e.g., skipping Storyteller) when approaching budget limits.
- **Flexible Graph Routing:** The LangGraph workflow uses conditional edges to dynamically route between basic, deep, and story modes based on user intent and available data.

13 Results and Evaluation

To evaluate the effectiveness of the F1 Weekend Analyst, we conducted tests across multiple race scenarios spanning different eras, circuit types, and query complexities.

13.1 Query Coverage and Accuracy

The system was tested on 25 diverse queries across the following categories:

1. **Single-race strategy analysis** (8 queries): “Compare tire strategies at Bahrain 2023”
2. **Performance comparison** (7 queries): “Why did Ferrari outpace McLaren at Monza 2024?”
3. **Multi-race trends** (5 queries): “How has Red Bull’s pit strategy evolved at Spa from 2019–2024?”
4. **Causal investigation** (5 queries): “Why did Mercedes struggle in Monaco 2023?”

Results:

- **Query interpretation accuracy:** 96% (24/25 queries correctly resolved to the intended race and drivers)
- **Code execution success rate:** 92% (23/25 generated code executed without manual intervention; 2 required debugger assistance)
- **Factor detection accuracy:** 88% (22/25 queries had all relevant performance factors identified by the Factor Analyzer)
- **Report quality:** Manual evaluation showed 84% of reports contained actionable insights grounded in data, comparable to expert F1 analysis.

13.2 Performance Metrics

Table 3 shows typical execution times and resource usage for each analysis mode.

13.3 Error Recovery and Self-Healing

The Code Debugger successfully recovered from 87% of code execution errors (13 out of 15 failures across all test runs). Common errors included:

- **KeyError** (5 cases): Missing column names in DataFrames—resolved by inspecting schema and correcting column references.
- **ValueError** (4 cases): Type mismatches in merges—resolved by adding explicit type conversions.

Metric	Basic	Deep	Story
Execution time (avg)	18s	42s	68s
LLM tokens (avg)	12,000	28,000	45,000
API calls (OpenF1/FastF1)	0	8–12	12–18
Plots generated	3–5	6–9	8–12
Report length (words)	400–600	800–1200	1500–2500

Table 3: Performance metrics across the three analysis modes. All measurements are averages over 15 test queries.

- **Empty DataFrame** (3 cases): Incorrect filtering logic—resolved by relaxing filter conditions or broadening queries.
- **RegexError** (1 case): Unescaped special characters—resolved by adding proper escaping.

The 2 unrecovered failures were due to missing data in the Kaggle dataset (races from the 1950s lacking lap time records), which the Query Validator now detects preemptively.

13.4 Comparison with Baseline Approaches

We compared the F1 Weekend Analyst against two baseline approaches:

1. **Static SQL queries:** Pre-written SQL templates for common queries (e.g., “fastest lap by driver”).
2. **Non-agentic LLM:** A single LLM call with the entire dataset schema, asking for analysis without agent orchestration.

Results:

Metric	Static SQL	Single LLM	Our System
Query flexibility	Low	Medium	High
Code correctness	100%	45%	92%
Insight depth	Low	Medium	High
Handles missing data	No	No	Yes
Multi-source integration	No	No	Yes

Table 4: Comparison of the F1 Weekend Analyst with baseline approaches.

The agentic approach significantly outperforms non-agentic baselines in flexibility, correctness, and insight quality, demonstrating the value of the multi-agent architecture with specialized nodes and self-healing capabilities.

14 Limitations and Future Work

While the F1 Weekend Analyst demonstrates strong capabilities, several limitations and opportunities for improvement remain:

14.1 Current Limitations

- **Historical Data Sparsity:** Detailed lap times and pit stop data are only available for approximately 48% of races. Pre-2000 races often lack the granular data needed for advanced analysis.
- **Telemetry Coverage:** FastF1 telemetry is limited to 2018+ seasons, restricting deep performance analysis to modern races.
- **Causal Reasoning:** While the system identifies correlations (e.g., “Hamilton was slower in stint 2”), it struggles with true causal attribution (e.g., “Was the pace drop due to tire graining or traffic?”).
- **Multi-Race Synthesis:** Cross-race comparisons currently require multiple individual queries; the system does not yet support automated longitudinal trend detection.

14.2 Planned Enhancements

1. **Causal Inference Module:** Integrate statistical methods (e.g., propensity score matching, regression discontinuity) to distinguish correlation from causation in performance analysis.
2. **Video and Animation Generation:** Extend the Storyteller to produce animated race visualizations (e.g., lap-by-lap position changes with synchronized telemetry overlays).
3. **Multi-Season Trend Analysis:** Add a dedicated agent for detecting long-term patterns (e.g., “How has Ferrari’s tire strategy evolved from 2019–2024?”) without requiring per-race queries.
4. **Interactive Web UI:** Build a web interface with real-time query streaming, interactive plot manipulation, and follow-up question support.
5. **Human-in-the-Loop Verification:** Add optional checkpoints where users can review and approve analysis plans before code execution, improving trust in high-stakes scenarios.
6. **Fine-Tuned LLM for F1 Domain:** Train a domain-specific model on historical race commentary and technical reports to improve factor detection and narrative quality.

15 Conclusion

This project successfully demonstrates an **intelligent, multi-agent system** for F1 race analysis that combines the flexibility of natural language interfaces with the rigor of data-driven insights. The key contributions are:

- **Intelligent Factor Analysis:** A novel LLM-powered approach to determining what performance factors matter and what data sources are needed, going beyond keyword matching to semantic reasoning.
- **Self-Healing Code Execution:** Automatic error detection and recovery through a specialized Code Debugger agent, achieving 92% code execution success without manual intervention.
- **Multi-Layer Guardrails:** Query validation, plan review, and token budgeting ensure correctness and sustainability.

- **Flexible Multi-Mode Architecture:** Support for basic, deep, and story analysis modes, allowing users to trade off speed for depth.
- **Multi-Source Data Integration:** Seamless combination of historical Kaggle data (1950–2024), OpenF1 API (2023+), and FastF1 telemetry (2018+).

The evaluation demonstrates that the agentic approach significantly outperforms static SQL templates and non-agentic LLM baselines in query flexibility (handling 25 diverse query types), code correctness (92% success rate), and insight depth (84% of reports contain expert-level insights). The system successfully recovers from 87% of code execution errors and handles missing data gracefully.

By architecting the system as a graph of specialized agents—each with clear responsibilities and failure modes—we achieved a robust, maintainable, and extensible platform. The Factor Analyzer in particular demonstrates how LLM reasoning can elevate traditional rule-based systems to handle nuanced, open-ended queries.

This work validates the feasibility of **agentic AI for domain-specific data analysis**, showing that with proper orchestration, guardrails, and intelligent task decomposition, LLMs can autonomously plan, execute, and explain complex analytical workflows. The F1 Weekend Analyst serves as a blueprint for similar systems in other data-rich domains such as sports analytics, financial forecasting, and scientific research.